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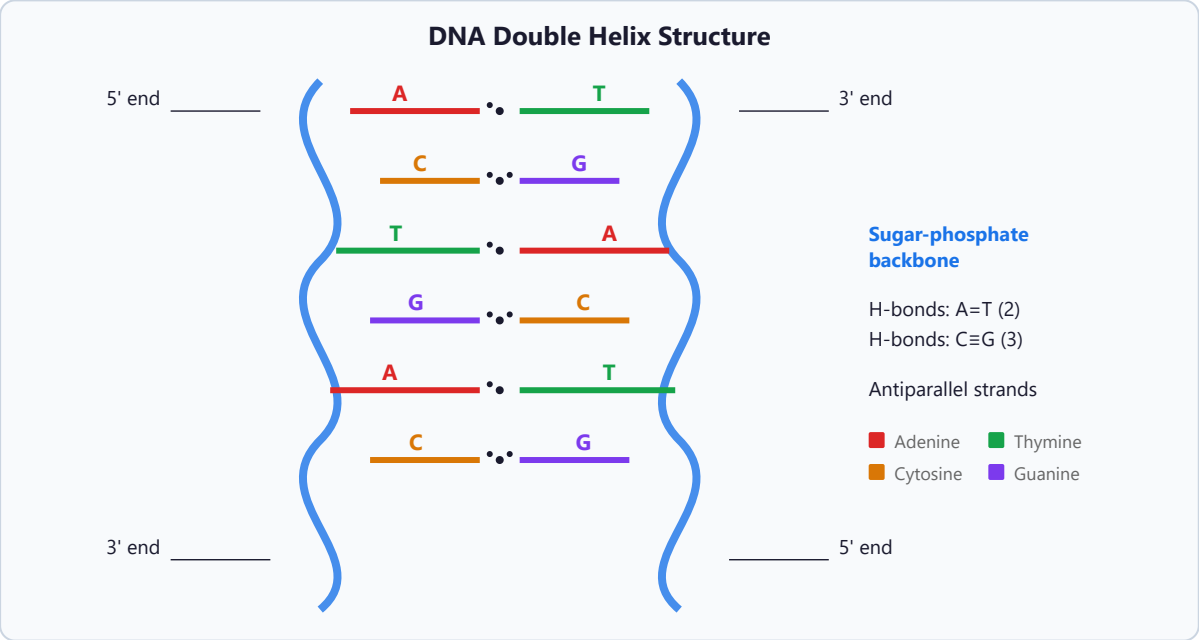
S2 — Genes and Health

2.1 DNA Structure and the Genetic Code

DNA Structure

DNA (deoxyribonucleic acid): A double-stranded polynucleotide that carries the genetic information of an organism. Each nucleotide consists of a deoxyribose sugar, a phosphate group, and one of four nitrogenous bases: adenine (A), thymine (T), cytosine (C), or guanine (G).



The two polynucleotide strands are held together by hydrogen bonds between complementary base pairs: adenine pairs with thymine (two hydrogen bonds) and cytosine pairs with guanine (three hydrogen bonds). The strands run antiparallel (one runs 5' to 3', the other 3' to 5') and are twisted into a right-handed double helix with approximately 10 base pairs per turn. The sugar-phosphate backbone provides structural stability (covalent phosphodiester bonds are strong) while the bases face inward carrying the genetic information. The two strands are complementary, meaning the sequence of one strand determines the sequence of the other — this is crucial for DNA replication and transcription.

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Nucleotide structure: Each nucleotide is formed by a condensation reaction joining a pentose sugar (deoxyribose in DNA, ribose in RNA), a phosphate group, and a nitrogenous base. Nucleotides are joined by phosphodiester bonds between the 3' carbon of one sugar and the 5' phosphate of the next, creating a directional backbone. Purines (adenine, guanine) have a double-ring structure; pyrimidines (thymine, cytosine) have a single ring. A purine always pairs with a pyrimidine, keeping the helix width constant.

Chargaff's rules: In any DNA molecule, the amount of adenine equals thymine, and the amount of cytosine equals guanine (A = T, C = G). This was key evidence supporting the Watson-Crick model of complementary base pairing. The ratio of A+T to C+G varies between species but is constant within a species.

Genes and the Genetic Code

> **Gene:** A sequence of DNA nucleotides that codes for a functional polypeptide (or functional RNA molecule). Genes are located at specific positions (loci) on chromosomes.

The genetic code has the following features:

- **Triplet:** Each codon consists of three consecutive bases coding for one amino acid.
- **Degenerate (redundant):** Most amino acids are coded for by more than one codon (there are 64 possible codons but only 20 amino acids).
- **Non-overlapping:** Each base is read only once, as part of only one codon.
- **Universal:** The same codons code for the same amino acids in almost all organisms (evidence of common ancestry).
- **Contains start and stop codons:** AUG is the start codon (also codes for methionine); UAA, UAG, and UGA are stop codons.

DNA Replication (Semi-Conservative)

DNA replication occurs during the S phase of the cell cycle and is described as **semi-conservative** because each new DNA molecule consists of one original (parent) strand and one newly synthesised strand. The process:

1. **Unwinding:** The enzyme **DNA helicase** unwinds the double helix and breaks the hydrogen bonds between complementary base pairs, separating the two strands and creating a replication fork.

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2. **Priming:** RNA primase synthesises a short RNA primer on each template strand, providing a free 3'-OH group for DNA polymerase to begin synthesis.
3. **Elongation: DNA polymerase** reads the template strand in the 3' to 5' direction and synthesises the new complementary strand in the 5' to 3' direction, adding free DNA nucleotides by complementary base pairing (A with T, C with G). The leading strand is synthesised continuously; the lagging strand is synthesised in short Okazaki fragments.
4. **Joining: DNA ligase** joins the Okazaki fragments on the lagging strand by forming phosphodiester bonds, creating a continuous strand.
5. **Proofreading:** DNA polymerase has 3' to 5' exonuclease activity, allowing it to remove incorrectly paired nucleotides and replace them, giving an error rate of approximately 1 in 10^9 bases.

Evidence for semi-conservative replication: The Meselson-Stahl experiment (1958) used *E. coli* grown in heavy nitrogen (^{15}N) medium, then transferred to light nitrogen (^{14}N) medium. After one round of replication, all DNA had intermediate density (one heavy and one light strand). After two rounds, half had intermediate density and half had light density. This conclusively supported semi-conservative replication over conservative or dispersive models.

> **Exam Tip:** When describing DNA replication, always name the enzymes (helicase, DNA polymerase, ligase) and state their specific roles. A common mistake is saying DNA polymerase "moves along" the template — be precise: it reads 3'→5' and synthesises 5'→3'. Remember that replication is bidirectional (two replication forks move in opposite directions from the origin).

RNA Structure

RNA differs from DNA in several ways: it is usually single-stranded (though it can fold into complex secondary structures via intramolecular base pairing), contains ribose sugar instead of deoxyribose (ribose has an -OH group on the 2' carbon), uses uracil (U) instead of thymine, and is generally shorter and less stable. The main types are:

- **Messenger RNA (mRNA):** Carries the genetic code from DNA in the nucleus to ribosomes in the cytoplasm. Its sequence of codons determines the amino acid sequence of the protein.
- **Transfer RNA (tRNA):** Small, clover-leaf shaped molecules that carry specific amino acids to the ribosome during translation. Each tRNA has an anticodon complementary to an mRNA codon and an amino acid attachment site at the 3' end.

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- **Ribosomal RNA (rRNA):** A structural and catalytic component of ribosomes, making up about 60% of ribosome mass. rRNA has peptidyl transferase activity, catalysing peptide bond formation.

2.2 Protein Synthesis

Transcription

Transcription occurs in the nucleus. The enzyme RNA polymerase binds to a promoter region on the DNA template strand and unwinds the double helix. RNA polymerase moves along the template strand in the 3' to 5' direction, synthesising a complementary mRNA strand in the 5' to 3' direction using free RNA nucleotides. Hydrogen bonds between DNA strands are broken ahead of the enzyme and reform behind it. When RNA polymerase reaches a terminator sequence, transcription stops and the pre-mRNA is released.

In eukaryotes, the pre-mRNA undergoes processing: introns (non-coding sequences) are spliced out and exons (coding sequences) are joined together. A 5' cap and 3' poly-A tail are added for stability and to aid export from the nucleus through nuclear pores.

Translation

Translation occurs at ribosomes in the cytoplasm. The mature mRNA binds to a ribosome. Transfer RNA (tRNA) molecules, each carrying a specific amino acid, have anticodons that are complementary to mRNA codons. The ribosome moves along the mRNA, reading codons sequentially. When a tRNA anticodon pairs with the mRNA codon in the ribosome's A site, the amino acid is joined to the growing polypeptide chain by a peptide bond (catalysed by peptidyl transferase in the ribosome). The tRNA then exits from the E site. Translation continues until a stop codon is reached, at which point a release factor binds and the polypeptide is released.

Exam Tip: Be precise about which strand is the template. The template strand is read 3' to 5'; the mRNA produced is complementary to it. The coding (sense) strand has the same sequence as the mRNA (with T instead of U). Always specify the direction of synthesis.

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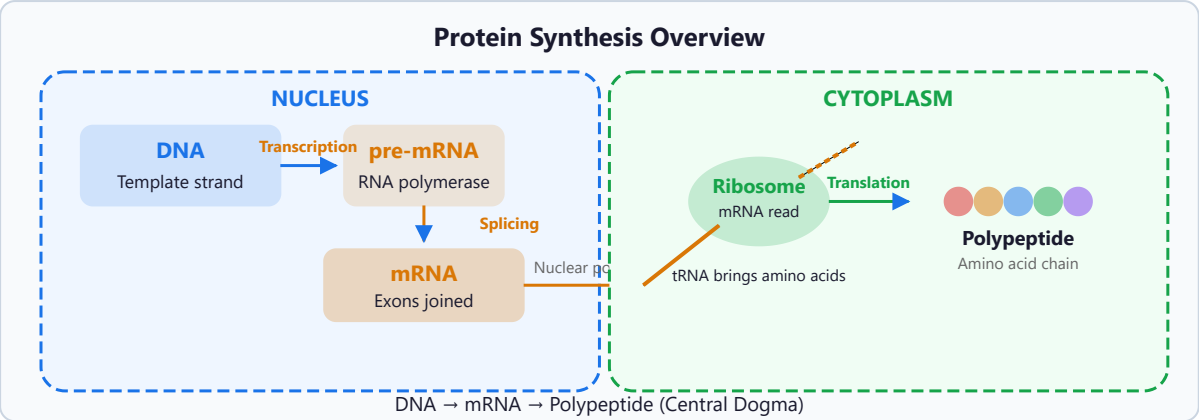


Post-Translational Modification

After translation, the polypeptide folds into its secondary structure (alpha helices and beta pleated sheets held by hydrogen bonds), tertiary structure (further folding stabilised by ionic bonds, disulfide bridges, hydrophobic interactions, and hydrogen bonds), and potentially quaternary structure (multiple polypeptide subunits). Some proteins require the addition of non-protein groups such as haem (in haemoglobin) or carbohydrate chains (glycoproteins).

2.3 Gene Mutations and Genetic Disease

Types of Gene Mutation

> **Gene mutation:** A change in the nucleotide sequence of DNA. This may be a substitution (one base replaced by another), insertion (an extra base added), or deletion (a base removed). Insertions and deletions cause a frameshift, altering every codon downstream of the mutation.

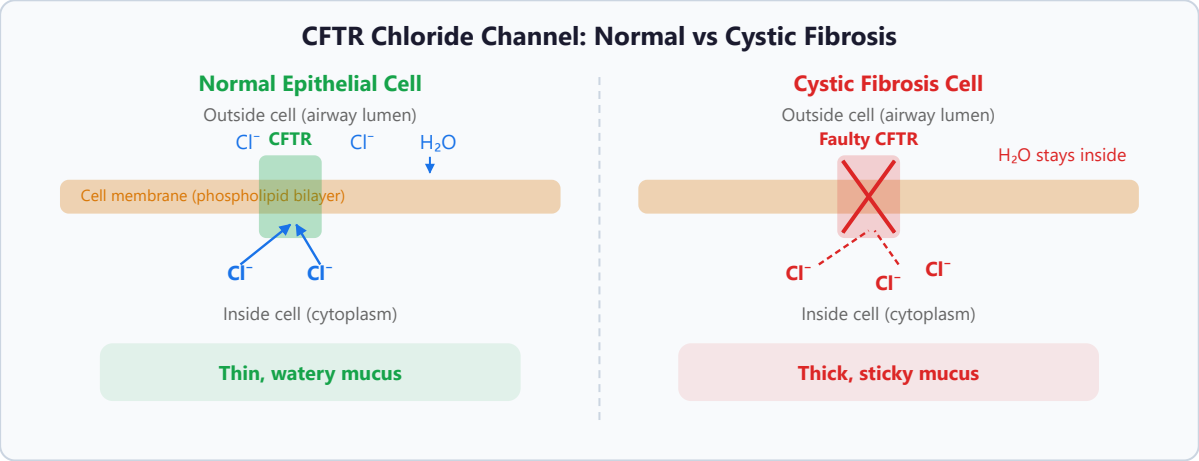
The consequences of mutations depend on their type and location:

- **Silent mutation:** The new codon still codes for the same amino acid (due to degeneracy of the code). No effect on the protein.
- **Missense mutation:** The new codon codes for a different amino acid. May or may not affect protein function depending on the nature of the change and its position.
- **Nonsense mutation:** The new codon is a stop codon, resulting in a truncated (shortened) and usually non-functional protein.

- **Frameshift mutation:** An insertion or deletion shifts the reading frame, changing all downstream amino acids. Almost always produces a non-functional protein.

Cystic Fibrosis

Cystic fibrosis (CF): An autosomal recessive genetic disorder caused by mutations in the CFTR gene on chromosome 7. The CFTR protein is a chloride ion channel in epithelial cell membranes. When it is absent or non-functional, chloride ions cannot be secreted out of cells, water does not follow by osmosis, and mucus becomes thick and sticky.



The most common mutation is ΔF508 — a deletion of three nucleotides that removes phenylalanine at position 508. This causes the CFTR protein to misfold so it is degraded before reaching the cell membrane. Over 2000 different mutations have been identified in the CFTR gene.

Symptoms of CF include:

- **Lungs:** Thick mucus clogs airways, reducing gas exchange surface area and trapping bacteria, leading to recurrent infections and progressive lung damage.
- **Pancreas:** Mucus blocks pancreatic ducts, preventing digestive enzymes from reaching the small intestine, causing malabsorption and malnutrition.
- **Reproductive system:** Thick cervical mucus in females reduces fertility; males often lack the vas deferens.
- **Sweat glands:** Produce abnormally salty sweat (used as a diagnostic test).

Genetic Inheritance of CF

CF is autosomal recessive, meaning an individual must inherit two copies of the faulty allele (be homozygous recessive, ff) to show the disease. Carriers (heterozygous, Ff) have one functional copy and one faulty copy — they produce enough functional CFTR protein to be unaffected but can pass the allele to offspring. If two carriers have a child, there is a 1 in 4 (25%) chance the child will have CF, a 1 in 2 (50%) chance of being a carrier, and a 1 in 4 chance of being homozygous dominant.

Genetic Cross: Two CF Carriers (Ff x Ff)

Father: **Ff** (carrier)
Gametes: F or f

Mother: **Ff** (carrier)
Gametes: F or f

	F	f
F	FF Unaffected	Ff Carrier
f	Ff Carrier	ff Has CF

25% FF | 50% Ff | 25% ff

Phenotype ratio = **3 unaffected : 1 CF**

F = normal CFTR allele (dominant) | f = faulty CFTR allele (recessive)

Exam Tip: In genetics questions, always define your allele symbols, draw a genetic diagram showing parental genotypes, gametes, a Punnett square, and offspring ratios. State the probability as a fraction, percentage, or ratio.

2.4 Cell Membranes

Fluid Mosaic Model

Fluid mosaic model: The accepted model of membrane structure in which a phospholipid bilayer forms a fluid matrix, with proteins (integral and peripheral), cholesterol, and glycolipids/glycoproteins scattered throughout in a mosaic pattern.

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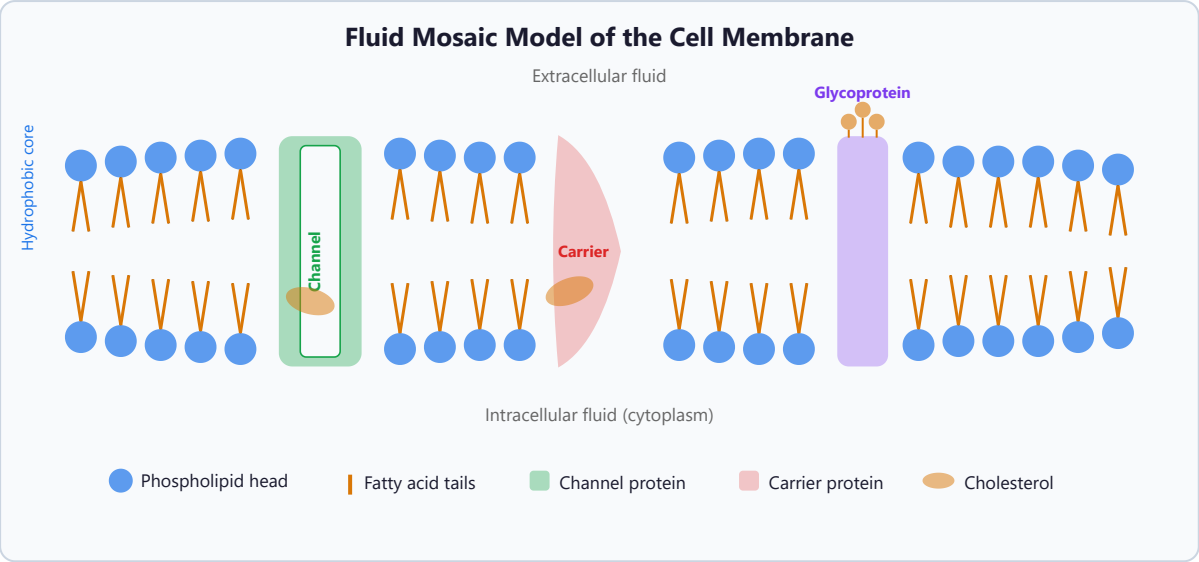

The phospholipid bilayer is the fundamental structure. Each phospholipid has a hydrophilic phosphate head and two hydrophobic fatty acid tails. In aqueous solution they spontaneously form a bilayer with heads facing outward and tails facing inward, creating a hydrophobic barrier to water-soluble substances.

Key components of the cell membrane:

Component	Function
Phospholipid bilayer	Forms the basic barrier; allows fat-soluble substances through; fluid structure allows membrane flexibility
Cholesterol	Regulates membrane fluidity — prevents phospholipids from packing too closely at low temperatures or moving too freely at high temperatures
Channel proteins	Provide hydrophilic pores for specific ions/polar molecules to pass through (passive transport)
Carrier proteins	Change shape to transport specific molecules across the membrane (facilitated diffusion or active transport)
Glycoproteins	Cell signalling, cell recognition, receptors for hormones and drugs
Glycolipids	Cell recognition and stability

Transport Across Membranes

Diffusion: The net movement of molecules or ions from a region of higher concentration to a region of lower concentration, down a concentration gradient. It is passive (requires no ATP) and continues until equilibrium is reached. Rate is affected by concentration gradient, temperature, surface area, and thickness of the membrane.

Facilitated diffusion: Diffusion of molecules or ions across a membrane via specific channel or carrier proteins. Still passive and moves down the concentration gradient, but allows hydrophilic or charged particles that cannot cross the phospholipid bilayer to pass through.

Osmosis: The net movement of water molecules from a region of higher water potential to a region of lower water potential across a partially permeable membrane. Water potential (ψ) is measured in kPa; pure water has the highest water potential (0 kPa).

Active transport: The movement of molecules or ions against their concentration gradient, from low concentration to high concentration, using ATP as an energy source. Requires carrier proteins that change shape when ATP is hydrolysed. Example: sodium-potassium pump (Na^+/K^+-ATPase) pumps 3 Na^+ out and 2 K^+ in per ATP molecule, maintaining the electrochemical gradient essential for nerve impulse transmission. Cells that carry out extensive active transport (e.g., root hair cells, kidney tubule cells, intestinal epithelial cells) have many mitochondria to supply the required ATP.

Endocytosis and exocytosis: Large molecules and particles that cannot cross the membrane via proteins are transported in vesicles. **Endocytosis** is the uptake of material into a cell by invagination of the membrane to form a vesicle (phagocytosis for solids, pinocytosis for liquids). **Exocytosis** is the reverse: vesicles fuse with the cell membrane to release their contents outside the cell (e.g., secretion of hormones, neurotransmitters, digestive enzymes). Both processes require ATP.

Exam Tip: When comparing transport mechanisms, organise your answer around key criteria: does it require energy (ATP)? Does it require a protein? Does it move down or against a concentration gradient? Is it specific to certain molecules? Diffusion: no energy, no protein, down gradient. Facilitated diffusion: no energy, protein required, down gradient. Active transport: energy required, protein required, against gradient. Osmosis: no energy, no protein (but requires partially permeable membrane), water only.

$$\text{Water potential } (\psi) = \text{Pressure potential } (\psi_p) + \text{Solute potential } (\psi_s)$$

2.5 Gas Exchange in the Lungs

Structure of the Gas Exchange System

The human gas exchange system consists of the trachea (reinforced with C-shaped cartilage rings), bronchi, bronchioles, and alveoli. The trachea and bronchi are lined with ciliated epithelium and goblet cells that produce mucus to trap pathogens and particulates; cilia beat rhythmically to move mucus up to the throat (the mucociliary escalator).

Alveoli and Gas Exchange

Gas exchange occurs across the alveolar surface by diffusion. Oxygen diffuses from the alveolar air into the blood in pulmonary capillaries, while carbon dioxide diffuses in the opposite direction. The alveoli are highly adapted for efficient gas exchange:

- **Large surface area:** ~70 m^2 provided by approximately 300 million alveoli.
- **Thin diffusion distance:** Alveolar wall is one cell thick, capillary wall is one cell thick — total diffusion distance is approximately 0.5 µm.
- **Rich blood supply:** Dense capillary network maintains a steep concentration gradient by continuously removing oxygen and delivering carbon dioxide.
- **Ventilation:** Breathing movements maintain a steep concentration gradient by replacing alveolar air with fresh atmospheric air.
- **Moist surface:** Gases dissolve in the thin layer of moisture lining the alveoli before diffusing across the membrane.

> **Fick's law of diffusion:** The rate of diffusion is proportional to (surface area × concentration difference) / thickness of the exchange surface. This summarises the adaptations of gas exchange surfaces.

$$\text{Rate of diffusion} \propto (\text{surface area} \times \Delta\text{concentration}) \,/\, \text{distance}$$

Gas Exchange and Cystic Fibrosis

In CF patients, thick sticky mucus accumulates in the airways because defective CFTR channels cannot secrete chloride ions, so water does not follow by osmosis to hydrate the mucus. This thick mucus: blocks

bronchioles, reducing ventilation of alveoli; is difficult for cilia to move, so the mucociliary escalator is impaired; traps bacteria (especially *Pseudomonas aeruginosa* and *Staphylococcus aureus*), leading to chronic infections and inflammation. Over time, repeated infections cause scarring and fibrosis of lung tissue, reducing the surface area for gas exchange and ultimately leading to respiratory failure.

CF Treatments

Current treatments manage symptoms rather than curing the underlying genetic defect:

- **Physiotherapy (chest percussion/postural drainage):** Loosens and drains thick mucus from the lungs.
- **Mucolytics (e.g., DNase/dornase alfa):** Enzyme that breaks down DNA released from dead neutrophils in the mucus, reducing its viscosity.
- **Antibiotics:** Used prophylactically and to treat acute lung infections. Nebulised antibiotics deliver drugs directly to the lungs.
- **Pancreatic enzyme supplements:** Taken with meals to replace the digestive enzymes that cannot reach the small intestine due to blocked pancreatic ducts.
- **Bronchodilators:** Widen airways to improve airflow.
- **CFTR modulators (e.g., ivacaftor, lumacaftor/ivacaftor, elexacaftor/tezacaftor/ivacaftor):** Newer drugs that target the defective CFTR protein. Some improve the function of misfolded CFTR protein at the cell surface; others help the protein fold correctly so it reaches the membrane. These represent a major breakthrough but only work for specific mutations.
- **Lung transplant:** May be necessary in severe cases when lung function deteriorates below a critical level.

Gene Therapy for CF

Gene therapy aims to insert a functional copy of the CFTR gene into the epithelial cells of the lungs. Approaches include:

- **Viral vectors (e.g., adenovirus, lentivirus):** Modified viruses deliver the gene efficiently, but may trigger an immune response, and the effect is temporary as epithelial cells are regularly replaced.
- **Liposome-mediated delivery:** The gene is wrapped in lipid vesicles (liposomes) that fuse with cell membranes. Safer than viral vectors but less efficient at gene delivery.

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- **Challenges:** The gene must reach enough cells to make a clinical difference; the thick mucus barrier is difficult to penetrate; expression is temporary; repeated treatments may be needed; immune responses to vectors; and the gene must be expressed at the correct level.

Ethical considerations: Gene therapy for CF is somatic (affects only the patient, not their offspring), which is generally considered ethically acceptable. Germline gene therapy (which would be heritable) raises greater ethical concerns and is currently prohibited in humans in most countries.

Exam Tip: When explaining how CF affects gas exchange, link the molecular defect (CFTR chloride channel) to the cellular effect (thick mucus) to the tissue/organ level effect (blocked airways, infections) to the whole organism effect (breathing difficulty, reduced oxygen delivery). This shows you can explain at multiple levels of organisation.

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